Po-Chien Luan

po-chien.luan@epfl.ch | Google Scholar | Website | Lausanne, Switzerland

RESEARCH SUMMARY

Ph.D. researcher at EPFL VITA Lab working at the intersection of world models, 3D computer vision, human motion prediction, and robot navigation. My research focuses on building predictive world models and socially-aware forecasting systems that enable embodied agents to understand dynamic environments, anticipate human behavior, and navigate robustly. I develop deep learning methods based on diffusion models, state-space models, monocular vision, multi-view geometry, and trajectory prediction, with applications to autonomous systems and human-centered robotics.

RESEARCH INTERESTS

- World Models • Human Motion • Social Navigation • State Space Models • 3D Computer Vision

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Sep. 2022 – Present
Ph.D. in Robotics Lausanne, Switzerland

- Advisor: Prof. Alexandre Alahi
- Collaboration with Honda R&D

National Cheng Kung University (NCKU) Sep. 2018 – Aug. 2020
M.S. in Electrical Engineering Taiwan

- Thesis: IFB-PSO for Optimized Realization of an Anthropomorphic Robotic Hand and Electromyographic Hand Gesture Classification Using Lifelong Learning
- Chinese Automatic Control Society 2020 Excellent Master Thesis Award
- Advisor: Prof. Tzuu-Hseng S. Li; Co-advisor: Prof. Ping-Huan Kuo

National Cheng Kung University (NCKU) Sep. 2014 – Jul. 2018
B.S. in Electrical Engineering Taiwan

SELECTED PUBLICATIONS

Drift-Resistant Navigation World Model with Anchored Epipolar Guidance
arXiv preprint, 2026
Po-Chien Luan, Zimin Xia, Wuyang Li, Yang Gao, Alexandre Alahi
Predicts sparse future anchors and uses bidirectional epipolar constraints to reduce perceptual and geometric drift in long-horizon visual prediction.
Links: Paper | Project

Social-Mamba: Socially-Aware Trajectory Forecasting with State-Space Models
arXiv preprint, 2026
Po-Chien Luan, Wuyang Li, Yang Gao, Alexandre Alahi
Structures agents into an ego-centric social grid and uses Cycle Mamba scans to model social interactions efficiently for trajectory forecasting.
Links: Paper | Code

Unified Human Localization and Trajectory Prediction with Monocular Vision
IEEE International Conference on Robotics and Automation (ICRA), 2025
Po-Chien Luan*, Yang Gao*, Céline Demonsant, Alexandre Alahi
Estimates human localization and future trajectories from a single camera by connecting 2D pose observations with trajectory prediction.
Links: Paper | Code

Sim-to-Real Causal Transfer: A Metric Learning Approach to Causally-Aware Interaction Representations
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
Ahmad Rahimi*, **Po-Chien Luan***, Yuejiang Liu*, Frano Rajič, Alexandre Alahi
Learns causally aware interaction representations in simulation and transfers them to real-world trajectory understanding.
Links: Paper | Code

Stable Video Infinity: Infinite-Length Video Generation with Error Recycling
International Conference on Learning Representations (ICLR), Oral, 2026
Wuyang Li, Wentao Pan, **Po-Chien Luan**, Yang Gao, Alexandre Alahi
Recycles prediction errors to maintain temporal stability and extend video synthesis to infinite-length generation.
Links: Paper | Project | Code

OmniTraj: Pre-Training on Heterogeneous Data for Adaptive and Zero-Shot Human Trajectory Prediction
CVPR Workshop on Precognition, 2026
Yang Gao, **Po-Chien Luan**, Kaouther Messaoud, Lan Feng, Alexandre Alahi
Pre-trains for adaptive and zero-shot human trajectory prediction across heterogeneous datasets, frame rates, input lengths, and unseen settings.
Links: Paper | Code

Multi-Transmotion: Pre-trained Model for Human Motion Prediction
Conference on Robot Learning (CoRL), 2024
Yang Gao, **Po-Chien Luan**, Alexandre Alahi
Learns transferable motion representations for forecasting future human movement in robotics contexts.
Links: Paper | Code

Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
Yang Gao, Wuyang Li, **Po-Chien Luan**, Alexandre Alahi
Models dynamic 3D scenes with deformable Gaussian occupancy and factorized foundation-model distillation for temporally consistent weakly supervised occupancy prediction.
Links: Paper | Code

WORK EXPERIENCE

Advanced Institute of Manufacturing with High-tech Innovations (AIM-HI) Jul. 2021 – May 2022
Graduate Research Assistant Taiwan

- Advisors: Prof. Her-Terng Yau; Prof. Ping-Huan Kuo
- Leading a team for smart manufacturing and supervising 6 master students
- Published 9 journal papers (3 first-author, 6 co-author)

Military Service Oct. 2020 – Mar. 2021
Taiwan

AWARDS AND HONORS

EPFL–Taiwan Ministry of Education (MoE) Scholarship (2022–2026); one of 5 awards annually
Excellent Teaching Assistant Award, Control Engineering Laboratory (NCKU EE) (2020)
FIRA RoboWorld Cup Androsot: 1st Place (2019)
FIRA RoboWorld Cup HuroCup (Adult Robots All-Round): 1st Place (2018)
KAIST AI World Cup (AI Soccer): 4th Place (2018)

ACADEMIC SERVICE

Teaching Assistantships
Lead Teaching Assistant, Deep Learning for Autonomous Vehicles, EPFL (2026)
Teaching Assistant, Deep Learning for Autonomous Vehicles, EPFL (2023–2024)
Teaching Assistant, Control Engineering Laboratory Course, NCKU (2019)

Reviewer
CVPR; ECCV; NeurIPS; RA-L;T-ITS; TRC; T-IE; T-II; T-CYB

SKILLS

Programming : Python, C/C++, MATLAB, Java
Libraries : PyTorch, Hugging Face, Weights & Biases (W&B), OpenCV
Development Tools : Git